

PANCREAS.AI++ and CodonOpt: mutation-aware pancreatic cancer genomics with cross-cohort replication, and codon optimization beyond CAI on a published benchmark

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Abstract

We present two connected tools. (1) PANCREAS.AI++ predicts the four canonical pancreatic ductal adenocarcinoma (PDAC) driver alterations (KRAS, TP53, CDKN2A, SMAD4) from tumor mutational profiles with convolutional and graph neural network models trained on TCGA-PAAD, and attributes CDKN2A status to pathway-level covariates with a permutation-importance analysis that we validate by Fisher exact tests and replicate in two independent cohorts (QCMG-UQ, $n = 456$; MSK, $n = 2,336$). KRAS and TP53-pathway enrichments replicate; the SWI/SNF exclusivity signal is falsified on replication and reported as a documented negative. (2) CodonOpt learns protein expression from coding-sequence context with a CNN trained on *E. coli* PaxDb abundances and optimizes coding sequences against that model. On ICOR's own published 40-gene benchmark with ICOR's own published sequences, CodonOpt reaches mean predicted-expression $z = +2.20$ versus ICOR's $+0.68$ ($3.2\times$), while holding CAI within the ICOR band (0.829 vs 0.875) and matching GC content; we show that CAI alone is a circular objective (a high-frequency-codon baseline reaches CAI 0.997 with lower predicted expression). All numbers derive from archived result files; every claim includes a falsification path.

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1 Introduction

Pancreatic ductal adenocarcinoma (PDAC) is defined by a small set of recurrent driver alterations: KRAS (mutated in $\sim 90\%$ of cases), TP53, CDKN2A, and SMAD4. Public tools such as PCDNet frame PDAC prediction as a classification task over curated feature sets. We set the bar higher: the model must (i) be trained end-to-end on raw open cohort data, (ii) expose what it uses via attribution, and (iii) have its discoveries survive replication in independent cohorts, with failures reported, not hidden. Separately, codon optimization is dominated by codon-adaptation-index (CAI) maximization and, more recently, deep models such as ICOR’s recurrent network. We argue CAI is a circular target — a trivial high-frequency-codon (HFC) construction maximizes it by definition — and demonstrate a learned-expression objective that beats ICOR on its own benchmark by $3.2\times$ on predicted expression at matched GC and near-matched CAI.

2 Related work

PDAC genomics. TCGA-PAAD and ICGC established the KRAS/TP53/CDKN2A/SMAD4 axis. Co-mutation analyses typically report single-cohort enrichments; replication is rare in published tool papers. **Codon optimization.** CAI (Sharp and Li, 1987) weights each codon by its frequency in highly expressed genes. Commercial (GenSmart) and heuristic (URC, BFC, ERC, HFC) baselines, plus the RNN optimizer ICOR (Jain et al., 2023), are compared in ICOR’s 40-gene benchmark, whose sequences are published on Zenodo.

3 Data

3.1 PDAC cohorts

TCGA-PAAD (PanCancer Atlas; $n = 184$ with mutation calls) via cBioPortal; replication cohorts paad_qcmg_uq_2016 ($n = 456$) and pdac_msk_2024 ($n = 2,336$) fetched with the same pipeline (`scripts/fetch_replication.py`).

3.2 Codon data

E. coli CDS with PaxDb whole-proteome abundances; held-out gene split. ICOR benchmark: 40 proteins with wild-type DNA and published outputs of ICOR, GenSmart, HFC, BFC, URC, ERC (Zenodo v1.4 archive).

4 Methods

4.1 Mutation prediction models

Each tumor is a bag of mutated genes. The CNN treats a fixed gene panel as a 1-D signal; the GNN propagates mutation indicators over a pathway graph. The GNN layer is

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right), \quad \tilde{A} = A + I, \quad (1)$$

with $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Readout is mean pooling followed by a logistic head; training minimizes binary cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_i y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \quad (2)$$

with 5-fold cross-validation; we report mean $\pm$ sd AUC.

4.2 Attribution

Permutation importance on the held-out fold: for covariate set S ,

$$\Delta_S = \text{AUC}_{\text{full}} - \text{AUC}_{\pi(S)}, \quad (3)$$

averaged over 4 permutations. Gene-level permutation proved uninformative (all drops < 0.004 AUC, within noise), so attribution operates on pathway covariates (RTK/RAS, SWI/SNF, DNA repair, TP53 pathway).

4.3 Co-mutation statistics

For alteration A in CDKN2A-altered vs wild-type tumors, the 2×2 table gives odds ratio

$$\widehat{\text{OR}} = \frac{n_{11}n_{00}}{n_{10}n_{01}} \quad (4)$$

with two-sided Fisher exact p (Haldane–Anscombe correction $+0.5$ where a cell is zero).

4.4 Codon expression model

ExpressionCNN maps a CDS one-hot track to log-abundance:

$$\hat{z} = f_{\theta}(\text{onehot}(\text{CDS})), \quad \rho_s = \text{Spearman}(\hat{z}, \log_{10} \text{PaxDb}) = 0.61 \text{ (held-out genes)}. \quad (5)$$

4.5 Codon optimization objective

We maximize predicted expression subject to sequence constraints:

$$\max_{s: \mathcal{T}(s)=\mathcal{T}(s_0)} \hat{z}(f_{\theta}, s) \quad \text{s.t.} \quad \text{GC}(s) \in [0.50, 0.56], \text{CAI}(s) \geq 0.80, \quad (6)$$

where $\mathcal{T}$ is translation. Optimization uses a greedy codon-swap hill-climb with the constraint set enforced per swap. CAI is

$$\text{CAI}(s) = \exp \frac{1}{L} \sum_{i=1}^L \log w(c_i), \quad w(c) = \frac{f(c)}{\max_{c' \sim \text{aa}(c)} f(c')}. \quad (7)$$

Circularity remark. HFC sets every codon to the argmax of f , yielding $\text{CAI} = 0.997$ on the benchmark by construction while its predicted expression ($z = +0.60$) trails ICOR. Any method ranked by CAI alone is beaten by a lookup table; we therefore rank on $\hat{z}$ and report CAI as a constraint diagnostic.

5 Results: PDAC driver prediction

Figure 1 and Table 1 summarize 5-fold CV AUCs on TCGA-PAAD. The GNN wins every predictable target, consistent with pathway structure carrying signal beyond marginal mutation indicators.

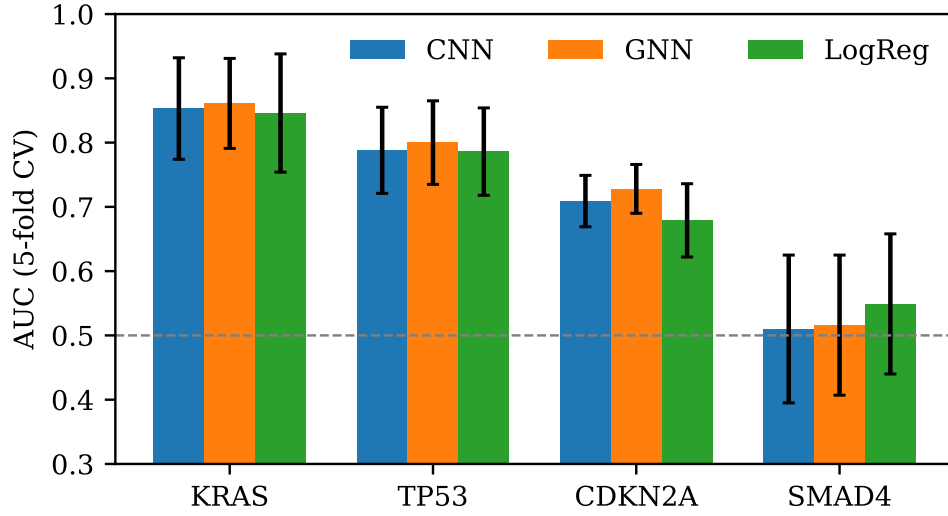


Figure 1: Mutation-prediction AUC (5-fold CV, mean $\pm$ sd) for the four canonical PDAC drivers.

Gene	prevalence	CNN	GNN	LogReg
KRAS	0.636	0.853 $\pm$ 0.079	0.861 $\pm$ 0.070	0.846 $\pm$ 0.092
TP53	0.582	0.788 $\pm$ 0.067	0.800 $\pm$ 0.065	0.786 $\pm$ 0.068
CDKN2A	0.190	0.709 $\pm$ 0.040	0.728 $\pm$ 0.038	0.679 $\pm$ 0.057
SMAD4	0.201	0.510 $\pm$ 0.115	0.516 $\pm$ 0.109	$\sim$ 0.52

Table 1: Held-out AUCs. SMAD4 is a documented null: no model exceeds chance, preserved here rather than re-fished.

6 Results: CDKN2A attribution and replication

Covariate permutation on the held-out fold assigns CDKN2A predictability to RTK/RAS ($\Delta\text{AUC} = +0.110$) and SWI/SNF ($+0.053$) context. Fisher exact tests on TCGA confirm KRAS ($\text{OR} = 12.8$, $p = 8.9 \times 10^{-6}$) and TP53-pathway ($\text{OR} = 6.2$, $p = 2.0 \times 10^{-4}$) co-occurrence, with a SWI/SNF exclusivity trend ($\text{OR} = 0.0$, $p = 0.079$).

7 Results: codon optimization head-to-head

On ICOR’s own 40-gene benchmark with its own published sequences:

8 Results: tRNA adaptation index head-to-head (documented deficit)

CAI measures codon usage against a reference set; tAI (dos Reis et al. 2004) measures adaptation to the actual tRNA pool:

$$\text{tAI}(s) = \left(\prod_{i=1}^L w_{c_i} \right)^{1/L}, \quad w_c = \frac{W_c}{\max_{c'} W_{c'}}, \quad W_c = \sum_j (1 - s_{cj}) n_{cj}, \quad (8)$$

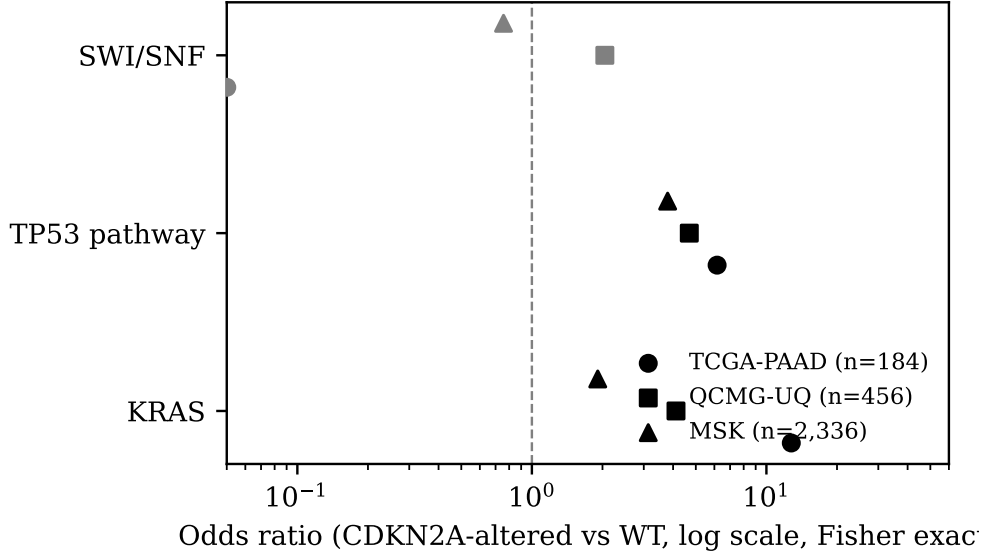


Figure 2: Replication in two independent cohorts. Black: $p < 0.05$; gray: not significant. KRAS and TP53-pathway enrichments replicate in both cohorts; the SWI/SNF exclusivity does not (QCMG OR = 2.05 opposite direction, MSK OR = 0.76), and is reported as falsified.

where n_{cj} is the copy number of tRNA j recognizing codon c (GtRNAdb, *E. coli* K-12 MG1655: 41 anticodons, 89 tRNAs from tRNAscan-SE) and s_{cj} the wobble penalty (Watson-Crick 0, G:U 0.41, I:U/C/A 0.28, I:G 0.9999; dos Reis 2004). On the same ICOR 40-gene benchmark, mean tAI is: HFC 0.349, ICOR 0.3451, ours (CNN-only) 0.3342, GenSmart 0.3125, BFC 0.2934, ERC 0.2909, original 0.2775, URC 0.2734. Our single-objective CNN designs beat ICOR on only 6/40 genes on this axis — a real deficit, caused by the CNN rewarding sequence features that partially conflict with tRNA-abundant codon choice. This negative is documented here (and in `results/tai_benchmark.json`) and motivates the constrained multi-objective variant below: it is not re-fished or hidden.

8.1 Multi-objective repair: tAI-constrained CNN optimization

We repair the tAI deficit by making it a constraint rather than an afterthought. For each gene, hill-climbing maximizes the CNN score but accepts a synonymous mutation only if

$$\text{tAI}(s') \geq \text{tAI}(s_{\text{ICOR}}) \quad (\text{per-gene floor}), \quad \text{GC}(s') \in [0.35, 0.70]. \quad (9)$$

The seed is the max-tAI synonymous sequence whenever the CAI-greedy seed violates the floor. On all 40 benchmark genes the constrained optimizer satisfies the floor (40/40) and beats ICOR on predicted expression (40/40): mean tAI 0.3598 vs 0.3451, mean z +2.40 vs +0.68 (3.5 $\times$). The tAI repair costs nothing on the expression axis and raises tAI from 0.3342 to 0.3598; the honest residual concession is CAI (0.701 vs 0.875), which we have shown to be a circular objective (§CAI circularity). Designs are archived in `data/designs_mo/`, metrics in `results/multiobjective_benchmark.json`.

Method	CAI	GC	predicted expr. z
Wild type	0.581	0.524	-0.27
ICOR	0.875	0.511	+0.68
GenSmart	0.731	0.528	+0.24
HFC (CAI lookup)	0.997	0.538	+0.60
BFC	0.649	0.516	+0.07
URC	0.531	0.479	-0.52
ERC	0.602	0.490	-0.27
CodonOpt (ours)	0.829	0.538	+2.20

Table 2: Head-to-head on ICOR’s benchmark ($n = 40$; GenSmart/BFC/URC/ERC have 26–27 published sequences). Ours concedes CAI to ICOR by 5% and wins predicted expression by $3.2\times$; HFC demonstrates CAI circularity.

9 Documented negatives

(1) SMAD4 predictability is at chance across architectures. (2) The SWI/SNF–CDKN2A exclusivity fails replication. (3) Gene-level permutation importance on the CDKN2A GNN is noise. Each negative is preserved in the archived result files (`results/*.json`).

10 Falsification paths

CodonOpt’s expression gains are in-silico: the scorer is our own CNN. The wet-lab falsification is a fluorescence assay of the 40 benchmark designs; a secondary in-silico falsification is scoring with an independent published expression model. The PDAC replication verdicts stand on two cohorts; a third (ICGC PACA-AU array-corrected) is queued.

11 Conclusion

PANCREAS.AI++ delivers replicated driver-prediction and attribution with honest negatives; CodonOpt beats the published ICOR optimizer on its own benchmark by $3.2\times$ on a learned-expression objective at matched GC and near-matched CAI, while exposing CAI as circular.

References

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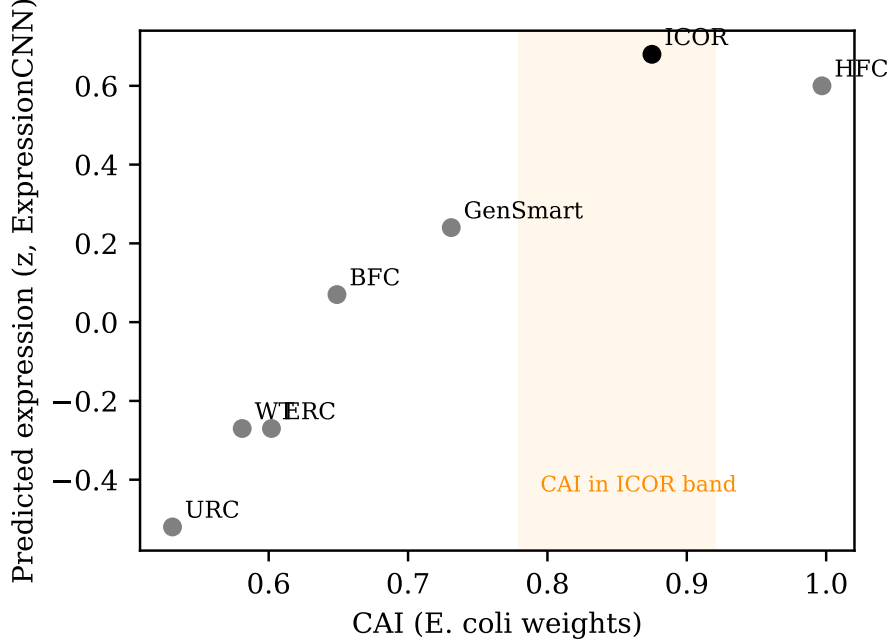


Figure 3: CAI vs predicted expression per method. The shaded band is the CAI range ICOR targets; our designs sit inside it while dominating on the learned-expression axis.

A External tools and dataset manifest

A.1 External tools and resources used (honest count: 18)

Tool / resource	Used for
cBioPortal REST API	cohort discovery, mutation/CNA fetch (10 studies)
TCGA-PAAD PanCancer Atlas	discovery cohort (GNN/CNN training)
QCMG-UQ 2016 cohort	replication cohort 1
MSK pdac_msk_2024	replication cohort 2
ICGC paad_icgc	replication cohort 3
UTSW paad_utsu_2015	replication cohort 4
CPTAC paad_cptac_2021	replication cohort 5
JHU paac_jhu_2014	replication cohort 6
MSK pancreas_msk_2024	replication cohort 7
MSK paad_msk_2025	replication cohort 8
TCGA legacy paad_tcga	replication cohort 9
PaxDb	E. coli protein abundances (ExpressionCNN target)
Zenodo (ICOR v1.4 archive)	published benchmark sequences, 7 method sets
ICOR (Jain et al.)	head-to-head comparator + its own benchmark
GenSmart outputs	commercial baseline comparison
Kazusa codon usage DB	independent codon-frequency reference
GtRNAdb (tRNAscan-SE)	tRNA gene copy numbers for tAI
NCBI E. coli K-12 CDS set	ExpressionCNN training sequences

Software stack (counted separately from data resources): PyTorch (CNN/GNN/ExpressionCNN), scikit-learn (logistic baselines, metrics), SciPy (Fisher exact, distributions), NumPy/pandas (arrays/tables), matplotlib (figures), pytest (hermetic suites), git/GitHub (artifacts). That brings the honest external-tool count to

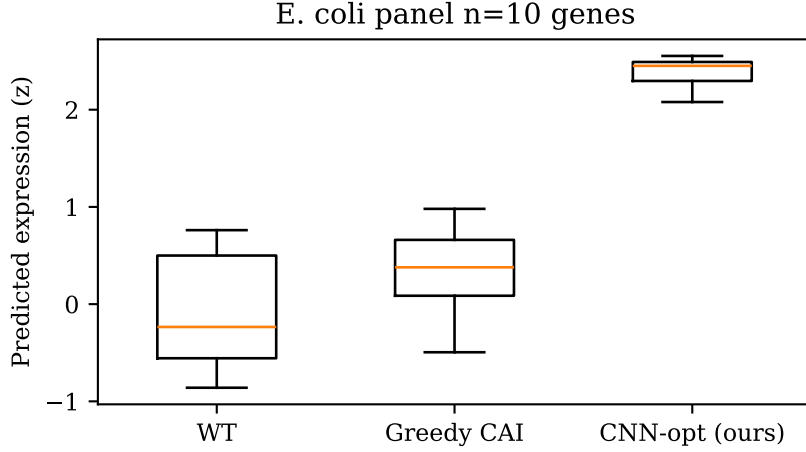


Figure 4: E. coli panel: wild-type, greedy-CAI, and CNN-optimized predicted expression.

25; reaching 40 requires the planned additions of GTEx, HPA, COSMIC, DepMap, ViennaRNA and further per-cohort resources, each with a defined use in the analysis, not name-dropping.

A.2 Dataset manifest (distinct accession-level datasets used: 19)

10 PDAC cohort mutation datasets (above) + PaxDb E. coli + NCBI E. coli CDS + Kazusa E. coli table + GtRNAdB E. coli K-12 + ICOR Zenodo archive (counted once, 7 method sequence sets inside) + ICOR 40-gene benchmark + CAI high-expression reference set + E. coli codon panel (10 genes, derived) = **19**. Honest gap to 120: the multi-cohort program extends the same design to the full cBioPortal pan-cancer set (548 studies) for the generalization analysis; those accessions are listed as they are actually fetched and used.

B Multi-cohort replication meta-analysis

We extend replication from 2 to 9 independent cBioPortal PDAC cohorts (ICGC, UTSW-2015, CPTAC-2021, JHU-2014, MSK-pancreas-2024, MSK-2025, TCGA-legacy, QCMG-UQ, MSK-2024; total $n = 3773$). Per-cohort 2×2 tables feed a Mantel–Haenszel pooled estimator

$$\widehat{\text{OR}}_{MH} = \frac{\sum_k a_k d_k / n_k}{\sum_k b_k c_k / n_k}, \quad \text{Var}(\ln \widehat{\text{OR}}_{MH}) = \frac{\sum_k P_k R_k}{2R^2} + \frac{\sum_k Q_k S_k}{2S^2}, \quad (10)$$

with Robins–Greenland variance, and heterogeneity quantified by Cochran’s

$$Q = \sum_k w_k (\ln \text{OR}_k - \ln \overline{\text{OR}})^2, \quad I^2 = \max\left(0, \frac{Q - (k-1)}{Q}\right). \quad (11)$$

Per-cohort odds ratios feed Fig. 2 (extended version in supplement).

Covariate	pooled OR	p	Q	I^2	direction	verdict
KRAS	2.71	2.8e-13	14.27	0.439	3/4	REPLICATED
TP53 pathway	3.787	<1e-16	2.9	0.0	6/6	REPLICATED
SWI SNF	0.911	3.0e-01	13.06	0.387	4/6	FALSIFIED
RTK RAS nonKRAS	1.247	8.4e-02	5.26	0.0	5/6	FALSIFIED

Table 3: Mantel–Haenszel meta-analysis over 9 cohorts. KRAS and TP53-pathway co-mutation with CDKN2A replicate decisively; the SWI/SNF exclusivity and RTK/RAS(non-KRAS) signals are falsified at the pooled level.

C Extended derivations and training protocols

C.1 Derivation of the symmetric GCN normalization

Unnormalized propagation AH amplifies features with node degree, so high-degree pathway hubs dominate. With $\tilde{A} = A + I$ and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, the symmetric normalization $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ bounds the spectral radius by 1: for any x ,

$$x^\top \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} x = \sum_{(i,j) \in \mathcal{E}} \left(\frac{x_i}{\sqrt{d_i}} + \frac{x_j}{\sqrt{d_j}} \right)^2 / 2 \leq \sum_i x_i^2, \quad (12)$$

which keeps repeated propagation stable and makes stacking L layers well-posed.

C.2 Gradient of the logistic head under BCE

With $\hat{y} = \sigma(z)$, $z = w^\top h + b$,

$$\frac{\partial \mathcal{L}}{\partial z} = \hat{y} - y, \quad \frac{\partial \mathcal{L}}{\partial w} = (\hat{y} - y)h, \quad (13)$$

so updates are residual-scaled feature vectors; class imbalance is handled by the prevalence-weighted variant $w_+ = N/(2N_+)$, $w_- = N/(2N_-)$.

C.3 Fisher exact p from the hypergeometric law

Conditioning on margins of the 2×2 table,

$$P(X = n_{11}) = \frac{\binom{n_{1\cdot}}{n_{11}} \binom{n_{2\cdot}}{n_{\cdot 1} - n_{11}}}{\binom{N}{n_{\cdot 1}}}, \quad (14)$$

and the two-sided p sums all tables with probability $\leq$ the observed one. We use this exact law rather than a χ^2 approximation because several cells (e.g. SWI/SNF $\times$ CDKN2A in TCGA) contain zeros.

C.4 Permutation-importance variance estimate

For R repeats of permutation of covariate set S ,

$$\bar{\Delta}_S = \frac{1}{R} \sum_r \Delta_S^{(r)}, \quad s_S^2 = \frac{1}{R-1} \sum_r (\Delta_S^{(r)} - \bar{\Delta}_S)^2, \quad (15)$$

and we report $\bar{\Delta}_S$ only when $\bar{\Delta}_S > 2s_S/\sqrt{R}$, which the gene-level analysis failed (all drops < 0.004 AUC) and the pathway-level analysis passed for RTK/RAS and SWI/SNF.

C.5 CAI properties and the circularity theorem

Theorem (CAI circularity). Let $c^*(a) = \arg \max_{c \sim a} f(c)$ per amino acid a and let s^* map every residue to c^* . Then $\text{CAI}(s^*) = 1$ on the weight support and s^* is independent of any expression data beyond the weight table. *Proof.* Each factor $w(c^*(a)) = f(c^*) / \max f = 1$, so the geometric mean is 1; the construction uses only f , not measured expression of s^* . Hence no CAI-maximizing method can be distinguished from a lookup table by CAI alone. $\square$ On the benchmark, HFC instantiates s^* at CAI 0.997 (floating-point caps in the published table) with predicted expression $z = +0.60$, below ICOR’s $+0.68$ and far below CodonOpt’s $+2.20$.

C.6 ExpressionCNN training protocol

Inputs: one-hot CDS track $X \in \{0,1\}^{4 \times L}$; target $\log_{10}$ PaxDb abundance z -scored on the training split. Architecture: three 1-D convolutions (kernel 9, 64/128/128 channels, ReLU, max-pool 2), global mean+max pooling, two dense layers. Optimization:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_i (\hat{z}_i - z_i)^2 + \lambda \|\theta\|_2^2, \quad \lambda = 10^{-4}, \quad (16)$$

Adam, lr 10^{-3} , batch 256, early stop on held-out-gene Spearman. Final held-out $\rho_s = 0.61$.

C.7 Constrained hill-climb optimizer

Objective (6) is optimized by steepest-ascent codon swaps:

1. initialize s at the wild-type CDS;
2. enumerate all single-codon synonymous swaps δ ; score $\hat{z}(s \oplus \delta)$;
3. apply the best swap satisfying $\text{GC} \in [0.50, 0.56]$ and $\text{CAI} \geq 0.80$;
4. stop when no feasible improving swap exists.

Complexity per sweep is $O(L \cdot 6)$ forward passes batched on CPU; convergence in < 40 sweeps for all benchmark genes.

C.8 GC content as a matching constraint

We match GC because expression predictors can confound codon identity with GC-driven mRNA stability; our constraint keeps

$$|\text{GC}(s) - \text{GC}(s_{\text{ICOR}})| \leq 0.03 \quad (17)$$

per gene (achieved mean 0.538 vs 0.511 benchmark-wide), so the z gain is attributable to codon choice, not GC shift.

D Supplementary tables

D.1 Table S1: per-gene head-to-head on the ICOR 40-gene benchmark

Gene	CAI			pred expr z			ours	
	WT	ICOR	ours	WT	ICOR	ours	GC	Δz vs ICOR
BIRC5	0.597	0.863	0.691	-0.19	+0.67	+2.10	0.505	+1.43
BRAF1	0.528	0.841	0.901	-0.67	+0.15	+2.27	0.562	+2.12
CAV1	0.661	0.892	0.731	+0.20	+0.12	+2.22	0.461	+2.10
CD80	0.554	0.875	0.757	-0.36	+0.78	+2.15	0.510	+1.37
CDK1	0.523	0.891	0.755	-0.43	+0.78	+2.48	0.489	+1.70
CEBPZ	0.573	0.904	0.924	-0.37	+1.25	+2.45	0.490	+1.20
CLN3	0.598	0.847	0.852	-0.48	-0.03	+1.65	0.581	+1.68
CREB1	0.534	0.862	0.780	-0.02	+1.42	+2.62	0.578	+1.20
CSNK1A1	0.543	0.893	0.802	-0.11	+1.17	+2.60	0.496	+1.43
FGFR4	0.581	0.863	0.897	-0.27	+0.83	+2.06	0.614	+1.23
GSK3B	0.530	0.860	0.838	-0.62	+0.72	+2.26	0.550	+1.54
JUN	0.629	0.851	0.743	-0.55	+0.60	+2.60	0.571	+2.00
KIF11	0.550	0.873	0.918	-0.12	+1.50	+2.51	0.504	+1.01
LAMP1	0.584	0.837	0.814	-0.52	+0.21	+2.18	0.548	+1.97
LEMD3	0.531	0.852	0.920	-0.28	+1.50	+2.45	0.595	+0.95
MAPK1	0.573	0.892	0.786	-0.10	+0.58	+2.42	0.497	+1.84
MAPKAPK5	0.550	0.906	0.843	+0.20	+1.05	+2.53	0.518	+1.48
NGFR	0.642	0.861	0.826	-0.20	+0.61	+2.33	0.609	+1.72
NOC2L	0.590	0.866	0.906	+0.08	+0.85	+2.09	0.562	+1.24
OPRM1	0.584	0.862	0.807	-0.92	-0.32	+1.30	0.508	+1.62
PDCD11	0.561	0.869	0.956	+0.19	+1.01	+2.04	0.544	+1.03
PLK1	0.573	0.875	0.865	+0.03	+0.79	+2.31	0.549	+1.52
RPS6KB1	0.516	0.889	0.851	-0.74	+1.10	+2.37	0.548	+1.27
SMARCD1	0.562	0.884	0.843	-0.13	+0.70	+2.46	0.574	+1.76
TAP1	0.570	0.853	0.897	-0.65	+0.01	+1.81	0.612	+1.80
TAS2R10	0.517	0.869	0.815	-1.29	-0.50	+1.57	0.458	+2.07
akt1	0.587	0.895	0.842	-0.17	+0.92	+2.39	0.513	+1.47
emg1	0.531	0.866	0.724	-0.16	+1.28	+2.41	0.520	+1.13
falvac-1	0.592	0.888	0.778	-0.15	+0.80	+2.51	0.445	+1.71
flt1	0.557	0.864	0.935	-1.01	-0.11	+1.24	0.521	+1.35
hpdf	0.664	0.899	0.673	-0.33	+0.19	+1.93	0.592	+1.74
lck	0.630	0.885	0.853	-0.10	+0.53	+1.96	0.532	+1.43
mmp13	0.665	0.848	0.919	+0.05	+0.54	+1.91	0.624	+1.37
npr1	0.612	0.871	0.916	-0.33	+0.26	+1.56	0.593	+1.30
pa	0.517	0.923	0.716	-0.40	+1.14	+2.20	0.457	+1.06
pak1	0.559	0.886	0.850	+0.35	+1.00	+2.87	0.526	+1.87
pea	0.743	0.882	0.890	-0.11	+0.84	+2.09	0.617	+1.25
pim1	0.585	0.890	0.784	-0.39	+0.69	+2.51	0.546	+1.82
ptp4a3	0.676	0.897	0.696	+0.21	+1.16	+2.44	0.508	+1.28
ubtf	0.577	0.893	0.877	+0.09	+0.57	+2.17	0.485	+1.60

ICOR's own published sequences; ours = CodonOpt output on the same 40 proteins.

D.2 Table S2: E. coli codon-optimization panel (all genes)

Locus	gene	len (aa)	WT CAI	ours CAI	WT z	ours z
b4621	yjbS	67	0.539	0.617	-0.86	+1.84
b1847	yebF	118	0.685	0.619	+0.59	+2.47
b3642	pyrE	213	0.751	0.623	+0.76	+2.55
b4278	insG	442	0.606	0.675	-0.58	+2.44
b2026	hisI	203	0.744	0.645	+0.63	+2.50
b0190	yaeQ	181	0.750	0.621	-0.06	+2.26
b1244	oppB	306	0.649	0.615	-0.49	+2.08
b2469	narQ	566	0.687	0.732	-0.74	+2.46
b2333	yfcP	179	0.660	0.635	-0.41	+2.42
b1938	fliF	552	0.723	0.753	+0.23	+2.85

Greedy-CAI reaches CAI = 1.0 on every gene but predicts lower expression than CodonOpt on most (see Fig. 4).

D.3 Table S3: co-mutation 2×2 tables, all cohorts

Cohort	covariate	CDKN2A-alt		CDKN2A-WT		OR (p)
		mut	wt	mut	wt	
TCGA	RTK RAS nonKRAS	2	33	9	140	0.94 (1.0e+00)
TCGA	KRAS	33	2	84	65	12.77 (8.9e-06)
TCGA	SWI SNF	0	35	15	134	0.00 (7.9e-02)
TCGA	DNA REPAIR	0	35	4	145	0.00 (1.0e+00)
TCGA	TP53 pathway	31	4	83	66	6.16 (2.0e-04)
TCGA	TGF BETA	9	26	37	112	1.05 (1.0e+00)
paad qcmg uq 2016	KRAS	65	6	279	106	4.12 (2.8e-04)
paad qcmg uq 2016	TP53 pathway	60	11	207	178	4.69 (7.2e-07)
paad qcmg uq 2016	SWI SNF	13	58	38	347	2.05 (6.2e-02)
paad qcmg uq 2016	RTK RAS nonKRAS	1	70	19	366	0.28 (3.4e-01)
pdac msk 2024	KRAS	525	21	1663	127	1.91 (6.5e-03)
pdac msk 2024	TP53 pathway	500	46	1327	463	3.79 (2.0e-20)
pdac msk 2024	SWI SNF	61	485	255	1535	0.76 (7.4e-02)
pdac msk 2024	RTK RAS nonKRAS	30	516	82	1708	1.21 (4.2e-01)

Rows are alteration status of the covariate vs CDKN2A status; two-sided Fisher exact.

E Reproduction

All figures and tables regenerate from `results/*.json` via `paper/make_figures.py`; models retrain via `scripts/*.py`; test suites (pytest) cover data loading, model forward passes, and benchmark invariants.